



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc CHEMISTRY

Programme	B. Sc. Chemistry				
Course Title	BIO CO-ORDINATION CHEMISTRY				
Type of Course	ELECTIVE IN MAJOR				
Semester	V				
Academic Level	300-399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	1. Understanding on Chemical bonding- Coordination bond 2. Interaction of ligands with metal ions 3. Brief idea on Bio-Inorganic Chemistry				
Course Summary	The course provides Understanding on the classification, significance, effects of ligands in biological systems. Analyze the interactions between ligands and metal ions, comprehend stability of complexes. Evaluate the functions and impacts of bulk metals (Na, K, Ca, Mg) in biological systems. Demonstrate knowledge of trace and ultratrace metals (Fe, Cu, Zn) roles in biological systems. Evaluate the applications of coordination compounds in medical therapy				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand classification and significance of ligands and summarise ligand effects in biological systems	U	F	Instructor-created exams / Quiz /Assignment
CO2	Analyze ligand interactions with metal ions in biological systems	An	C	Class test /Assignment /Quiz
CO3	Evaluate the functions and impacts of bulk metals (Na, K, Ca, Mg) in biological systems	E	P	Assignment/ Class test
CO4	Demonstrate knowledge of trace and ultratrace metals (Fe, Cu, Zn) roles in biological systems	An	P	Assignments /Seminar presentation

CO5	Evaluate the applications of co-ordination compounds in medical therapy	E	P	Assignments /Seminar presentation
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* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C)
 # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I	LIGANDS IN BIOLOGICAL SYSTEMS		12	20
	1	Classification of ligands	1	
	2	Biologically significant ligands- H ₂ O, NH ₃ , Amino acids, peptides, proteins and DNA- RNA bases	4	
	3	Chelate effect and Macrocyclic effect	2	
	4	Significance of Chelate and Macrocyclic and macrobicyclic effects in biological systems.	5	
II	LIGAND INTERACTIONS IN BIOLOGICAL SYSTEMS		12	20
	5	Classification of metal ions into bulk, trace and ultra-trace elements	1	
	6	Concept of hard and soft acids and bases	2	
	7	Selectivity of ligands to different metals	2	
	8	Essentials of Crystal-Field splitting	2	
	9	Stability of complexes- thermodynamic and kinetic stability. Stability constant. Factors affecting stability of metal complexes, metal centered and ligand centered properties	4	
	10	Inert and Labile complexes	1	
III	ROLE OF BULK METALS IN BIOLOGICAL SYSTEMS: (Na, K, Ca and Mg)		12	20
	11	Ionophores and its classification. Selectivity of ionophores.	1	
	12	Active and passive transport	1	
	13	Sodium potassium pump-mechanism- enzymes responsible for Na-K pump	2	
	14	Potassium deficiency and K excess effects in biological systems.	1	

	15	Structural role of Calcium in Muscle contraction, bone management and teeth management (qualitative).	3	
	16	Role of Ca in blood clotting, Storage and transfer of Calcium (Brief idea)	2	
	17	Role of Magnesium- structural and functional role in biological systems.	2	
IV	ROLE OF TRACE AND ULTRATRACE METALS IN BIOLOGICAL SYSTEMS (Fe, Cu and Zn)		12	20
	18	Oxygen management Fe proteins (Haemoglobin and myoglobin), Metal management Fe proteins (ferritin and transferrin), electron management Fe proteins (cytochromes and Fe-S proteins)	3	
	19	Oxygen management Cu proteins (Hemocyanin), Metal management Cu proteins (ceruloplasmin), electron management Cu proteins (cytochromes and plastocyanin)	3	
	20	Biological role of Zn- Lewis acids role, structural role, and functional role- Zinc enzymes (carbonic anhydrase)	3	
	21	Biological role of cobalt- Vitamin B12 co-enzymes	1	
	22	Bioinorganic aspects of Photosynthesis and nitrogen fixation (Brief discussion). Nitrogenase enzyme (qualitative)	2	
V	APPLICATION OF COORDINATION COMPOUNDS IN MEDICINE AND THERAPY- OPEN ENDED		12	20
	23	Arthritis drugs (Gold based), Diabetic drugs (Vanadium based). Chelation therapy – Use of Dimercapto propanol. Chemotherapy- Cis platin and new generation Pt drugs- Drug resistance and DNA repair mechanism of Pt drugs.	12	

References

1. J. E. Huheey, E. A. Keiter, R. L. Keiter, O. K. Medhi, Inorganic Chemistry- Principles of structure and reactivity, Pearson education.
2. D.E. Fenton, Bi- Coordination Chemistry, Oxford, 1995
3. D.F. Shriver and P.W. Atkins, Inorganic Chemistry, Oxford University Press
4. Rosette M. Roat- Malone- Bioinorganic Chemistry-A short Course- John Wiley & Sons, INC., Publication.

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PS O5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	1						3				2		1
CO 2	2						3				3		2
CO 3	3				1		3						
CO 4					2	1	3				3		3
CO 5						1	3				2		3

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory/Practical Exam
- Assignments /Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory/Practical Exam	Assignment / Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓